

Computer Networks — Reference Handbook

Computer Networks (CN) · Unit 1 · Introduction & Physical Layer · 2024

Overview

These notes cover **Computer Networks**. From bits on the wire to HTTP — layered network architectures, routing, transport protocols and the applications on top. Each section states the core idea, a worked intuition and the points examiners look for.

1. OSI Reference Model

The OSI model has seven layers: physical, data link, network, transport, session, presentation and application.

Why it matters: OSI Reference Model appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of osi reference model in one line.
- A small example showing osi reference model in practice.
- Common mistakes students make and how to avoid them.

2. TCP/IP Model

The TCP/IP stack collapses OSI into link, internet, transport and application layers.

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3. Network Topologies

Bus, star, ring, mesh and hybrid topologies differ in cost, fault tolerance and performance.

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- Key definition of network topologies in one line.
- A small example showing network topologies in practice.
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4. Framing

Character count, byte stuffing and bit stuffing mark frame boundaries.

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- Key definition of framing in one line.
- A small example showing framing in practice.
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5. Error Detection

Parity, checksums and CRC detect transmission errors.

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- Key definition of error detection in one line.
- A small example showing error detection in practice.
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6. Hamming Code

Hamming codes add parity bits that can locate and correct single-bit errors.

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- Key definition of hamming code in one line.
- A small example showing hamming code in practice.
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7. IPv4 Addressing

IPv4 addresses are 32-bit, organised into classes or CIDR blocks.

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8. Subnetting

Subnet masks split an address block into smaller networks for efficient allocation.

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9. Distance Vector Routing

Routers share distance vectors with neighbours using the Bellman-Ford equation; count-to-infinity is a known issue.

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10. UDP

UDP is a connectionless, best-effort transport with minimal header overhead.

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- Key definition of udp in one line.
- A small example showing udp in practice.
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11. TCP Connection Management

TCP opens connections with a three-way handshake and closes them with a four-way exchange.

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- A small example showing tcp connection management in practice.
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12. TCP Flow Control

The receive window prevents a fast sender from overwhelming a slow receiver.

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13. DNS

DNS resolves domain names via a hierarchy of root, TLD and authoritative servers.

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- Key definition of dns in one line.
- A small example showing dns in practice.
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14. HTTP

HTTP is a stateless request-response protocol; persistent connections and caching improve performance.

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- A small example showing http in practice.
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15. Email Protocols

SMTP sends mail while POP3 and IMAP retrieve it.

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- Common mistakes students make and how to avoid them.

Quick revision

OSI Reference Model · TCP/IP Model · Network Topologies · Framing · Error Detection · Hamming Code · IPv4 Addressing · Subnetting · Distance Vector Routing · UDP · TCP Connection Management · TCP Flow Control · DNS · HTTP · Email Protocols